

could just use derivations from lecture!

$$V_p = 0, \quad V_d = V_p - V_n = -V_n, \quad V_o = -aV_n$$

$$i_1 = \frac{V_i - V_n}{R_1} = \frac{V_i + V_o/a}{R_1} = i_2 = \frac{V_n - V_o}{R_2} = -\frac{V_o/a - V_o}{R_2}$$

$$-\frac{R_2}{R_1} \left(V_i + \frac{V_o}{a} \right) = V_o \left(1 + \frac{1}{a} \right)$$

$$-\frac{R_2}{R_1} V_i = V_o \left(1 + \frac{1}{a} + \frac{R_2}{R_1 a} \right) = V_o \left(1 + \frac{R_1 + R_2}{R_1 a} \right)$$

$$\frac{V_o}{V_i} = -\frac{R_2}{R_1} \left[\frac{1}{1 + \frac{R_1 + R_2}{R_1 a}} \right] \quad A_{ideal} = -\frac{R_2}{R_1}$$

$$R_1 = 10k\Omega, \quad R_2 = 400k\Omega \rightarrow A_{ideal} = -40V/V$$

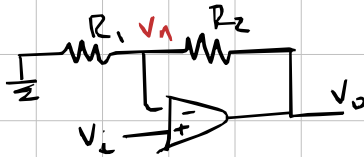
10% error! $10\% = \frac{A_{ideal} - A}{A_{ideal}} \rightarrow A = A_{ideal} (90\%)$
 $A = -36V/V$

$$-36V/V = -40V/V \left[\frac{1}{1 + \frac{10+400}{10a}} \right]$$

solve for $a \rightarrow$

$$a = 369$$

②



$$V_d = V_- - V_n, \quad V_o = a(V_- - V_n)$$

$$V_n = V_o \frac{R_1}{R_1 + R_2} \rightarrow V_o = a \left(V_i - V_o \frac{R_1}{R_1 + R_2} \right)$$

$$\frac{V_o}{a} \left(\frac{R_1 + R_2}{R_1} \right) = \left(\frac{R_1 + R_2}{R_1} \right) V_i - V_o$$

$$V_o \left(1 + \frac{1}{a} \frac{R_1 + R_2}{R_1} \right) = \frac{R_1 + R_2}{R_1} V_i$$

$$\frac{V_o}{V_i} = \left(1 + \frac{R_2}{R_1} \right) \left[\frac{1}{1 + \frac{1}{a} \left(1 + \frac{R_2}{R_1} \right)} \right] \quad A_{ideal} = 1 + \frac{R_2}{R_1}$$

$$R_1 = 10k\Omega, \quad R_2 = 400k\Omega \rightarrow A_{ideal} = 41 \text{ V/V}$$

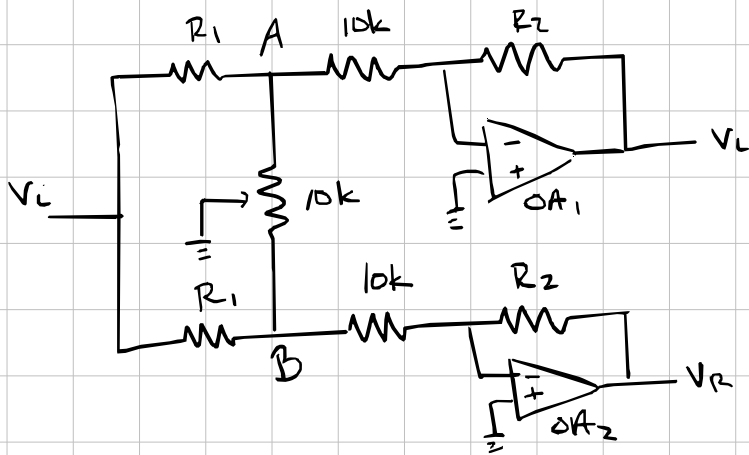
$$10\% \text{ error! } A = 41 \cdot 0.9 = 36.9 \text{ V/V}$$

$$36.9 = 41 \left[\frac{1}{1 + \frac{1}{a} (41)} \right]$$

$$\rightarrow a = 369\%$$

No difference from 1!

③ Book 1.27. $a = \infty$



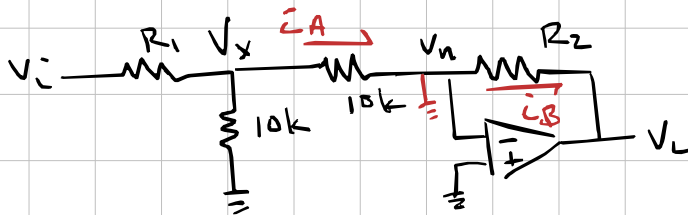
a) The left (top) and right (bottom) audio channels are essentially just inverting amplifiers with the input attenuated by the potentiometer (panpot) connecting the top and bottom. However, since the wiper is grounded, the top and bottom are independent and can be analyzed one at a time. Further, they are the same, so analysis for the top applies to the analysis of the bottom.

When the pot wiper is fully up, node A is grounded and the input has no effect on VL. (Node B connected to 10k resistor to ground)

When the pot wiper is fully down, node A is connected to ground via a 10k resistor, providing the least amount of attenuation. (Node B grounded)

When in middle, both node A and node B attenuated with a 5k resistor to ground

b) fully down / fully up cases the same, so consider only 1: $V_L/V_I = -1V/V$



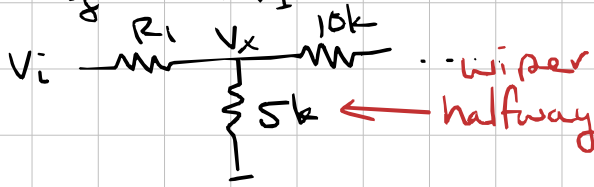
$$a = \infty \rightarrow V_N = 0V$$

$$V_X = \frac{10k \parallel 10k}{R_1 + 10k \parallel 10k} V_I = \frac{5k}{R_1 + 5k} V_I$$

$$i_A = i_B \rightarrow \frac{V_X}{10k} = -\frac{V_L}{R_2} = \frac{5k/R_1 + 5k}{10k} V_I = -\frac{V_L}{R_2}$$

$$V_L/V_I = -1V/V \rightarrow \frac{5k/R_1 + 5k}{10k} = \frac{1}{R_2} \rightarrow R_2 = 2(R_1 + 5k) \quad (1)$$

Midway: $V_L/V_I = -1/52$



$$V_X = \frac{5k \parallel 10k}{R_1 + 5k \parallel 10k} V_I \rightarrow \frac{3.3k}{R_1 + 3.3k} V_I = -\frac{V_L}{R_2}$$

$$V_L/V_I = -0.707 \rightarrow R_2 = 2.121(R_1 + 3.3k) \quad (2)$$

solving (1) & (2) $\rightarrow$

$$\begin{aligned} R_1 &= 24.2k\Omega \\ R_2 &= 58.7k\Omega \end{aligned}$$

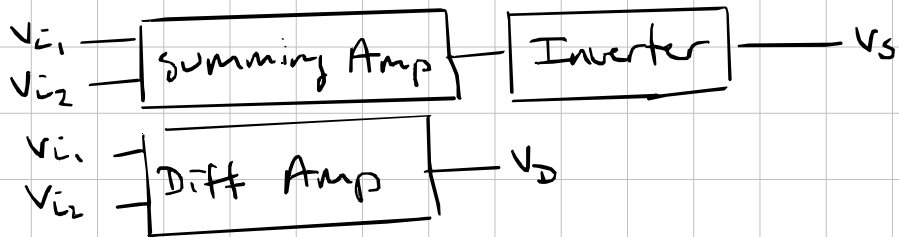
④ Book 1.37 $a = \infty$ OTHER SOLUTIONS POSSIBLE

with V_{i1} & V_{i2} , want $V_S = V_{i1} + V_{i2}$ & $V_D = V_{i1} - V_{i2}$

(inverting) summing Amp gives $V_S = -(V_{i1} + V_{i2})$ (unity gain)

Difference amp gives: $V_D = V_{i1} - V_{i2}$ (unity gain)

Simple 3 Amp solution: (1/2 credit)

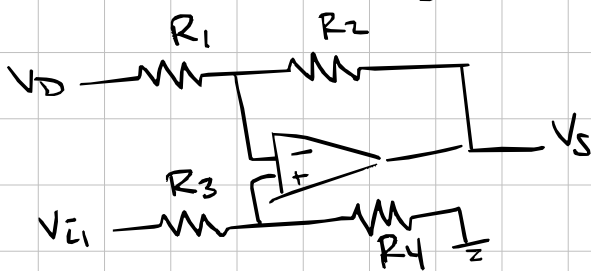


2 Amp Solution:

Use V_D , V_{i1} , V_{i2} to produce $V_S = V_{i1} + V_{i2}$

$$2V_{i1} - V_D = 2V_{i1} - (V_S - V_{i2}) = V_{i1} + V_{i2}$$

Above generated by another Difference amp!



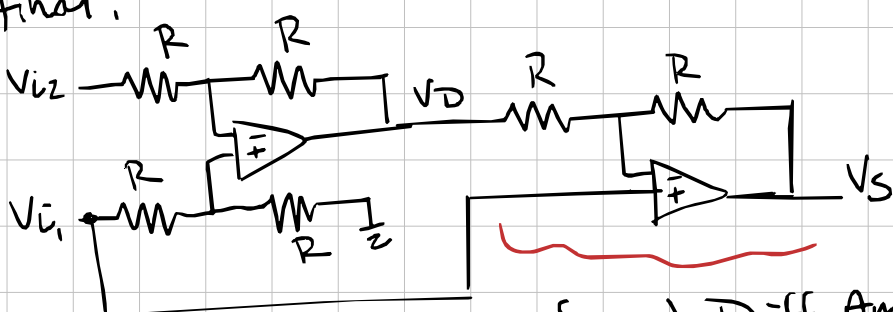
$$V_S = \frac{R_2}{R_1} \left(\frac{1 + R_1/R_2}{1 + R_3/R_4} V_{i1} - V_D \right)$$

want unity gain on V_D $\therefore \frac{R_2}{R_1} = 1 \rightarrow R_2 = R_1$

want $\times 2$ on V_{i1} $\therefore \frac{1 + R_1/R_2}{1 + R_3/R_4} = 2 = \frac{1 + 1}{1 + R_3/R_4} = \frac{2}{1 + R_3/R_4}$

$$2 = \frac{2}{1 + R_3/R_4} \rightarrow \frac{R_3}{R_4} = 0 \text{ let } R_3 = 0, R_4 \text{ doesn't matter, so } R_4 = \infty$$

final:



Second Diff Amp just a non-inverting Amp with V_D as Bias!

⑤ $a_2 = 50^3 \text{ V}$, $A = 40$

i) $A = \frac{1}{b} \left[\frac{1}{1 + \frac{1}{a_2 b}} \right] = \frac{a_2}{1 + a_2 b}$

$$1 + a_2 b = a_2 / A$$

$$b = \left[\frac{a_2}{A} - 1 \right] / a_2 \rightarrow b = 0.024992$$

$$\therefore A_{\text{ideal}} = 40.013$$

$$L = 50^3 (0.024992) = 3124$$

Effectively:

$$b = 0.025$$

$$A_{\text{ideal}} = 40$$

ii) a_2 decrease by 5%

Assume small change, $\frac{\Delta A}{A} = \frac{1}{1+L} \frac{\Delta a_2}{a_2} = \frac{1}{1+L} (0.05)$ ↑ 5%

$$\Delta A / A = \frac{1}{3125} 0.05 = 0.000016 = 0.0016\%$$

$$A = 40 (1 - 0.000016) = 39.9994$$

a_2 decreases by 50% → Not small!
from before, $b = 0.024992$

$$a_2 = 50^3 / 2 = 62500$$

$$A = \frac{1}{b} \left[\frac{1}{1 + \frac{1}{a_2 b}} \right] = 40.013 \left[\frac{1}{1 + \frac{1}{62500 \cdot 0.024992}} \right]$$

$$A = 39.987 \text{ V}, \text{ a } 0.03\% \text{ drop!}$$

A extremely insensitive to a_2 changes.

iii) Repeat but with a_2 starting as 50^2 V

$$b = 0.0246 \quad (A_{\text{ideal}} = 40.650)$$

$$L = 61.5$$

$$a_2 \text{ 5\% drop} \rightarrow \frac{\Delta A}{A} = \frac{1}{62.5} 0.05 = 0.08\%$$

$$A = 40 (1 - 0.0008) = 39.968$$

a_2 50% drop

$$A = \frac{a_2}{1 + a_2 b} \text{ with } a_2 = 50^2 (1 - .5) = 1250$$

$$A = 39.370, \text{ a drop of } 1.5\%$$

more sensitive!

⑤ $a_2 = 50 \text{ kV/V}$, $A = 40$

i) $A = \frac{1}{b} \left[\frac{1}{1 + \frac{1}{a_2 b}} \right] = \frac{a_2}{1 + a_2 b}$

$$1 + a_2 b = a_2 / A$$

$$b = \left[\frac{a_2}{A} - 1 \right] / a_2 \rightarrow b = 0.02498$$

$$\therefore A_{\text{ideal}} = 40.032$$

$$L = 50 \text{ k} (0.02498) = 1249$$

Effectively:

$$b = 0.025$$

$$A_{\text{ideal}} = 40$$

ii) a_2 decrease by 50%

Assume small change, $\frac{\Delta A}{A} = \frac{1}{1+L} \frac{\Delta a_2}{a_2} = \frac{1}{1+L} (0.05)$ ↑ 5%

$$\Delta A / A = \frac{1}{1250} 0.05 = 0.00004 = 0.004\%$$

$$A = 40 (1 - 0.00004) = 39.998$$

a_2 decreases by 50% → Not small!
from before, $b = 0.02498$

$$a_2 = \frac{1}{2} 50 \text{ k} = 25 \text{ k}$$

$$A = \frac{1}{b} \left[\frac{1}{1 + \frac{1}{a_2 b}} \right] = 40.032 \left[\frac{1}{1 + \frac{1}{25 \text{ k} \cdot 0.02498}} \right]$$

$$A = 39.968 \text{ V/V}, \text{ a } 0.08\% \text{ drop!}$$

A extremely insensitive to a_2 changes.

iii) Repeat but with a_2 starting as 5 kV/V (50×10^2)

$$b = 0.0248 \quad (A_{\text{ideal}} = 40.323)$$

$$L = 124$$

$$a_2 \text{ 5\% drop} \rightarrow \frac{\Delta A}{A} = \frac{1}{1+124} 0.05 = 0.04\%$$

$$A = 40 (1 - 0.0004) = 39.984$$

a_2 50% drop

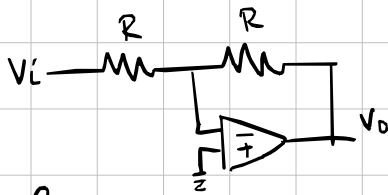
$$A = \frac{a_2}{1 + a_2 b} \text{ with } a_2 = 5 \text{ k} (1 - 0.5) = 2.5 \text{ k}$$

$$A = 39.683, \text{ a drop of } 0.79\%$$

more sensitive!

⑥ Book 1.49

$$V_i = 5 \sin(2\pi ft), \quad f = 1 \text{ kHz}$$



$$a = \begin{cases} 10 \text{ V/mV} & V_o > 0 \\ 2.5 \text{ V/mV} & V_o < 0 \end{cases}$$

for an inverting amp, lecture 1 showed:

$$a_c = a \frac{R_2}{R_1 + R_2} = -0.5a$$

$$A_{ideal} = -\frac{R}{R} = -1 \rightarrow b = -1$$

since inverting if

$$V_i < 0, V_o > 0, a = 10 \text{ V/mV} = 10000 \text{ V/V}$$

$$L = -0.5(10000)(-1) = 5000$$

$$A = A_{ideal} / (1 + 1/L) = -0.9998 \quad \text{Effectively } -1!$$

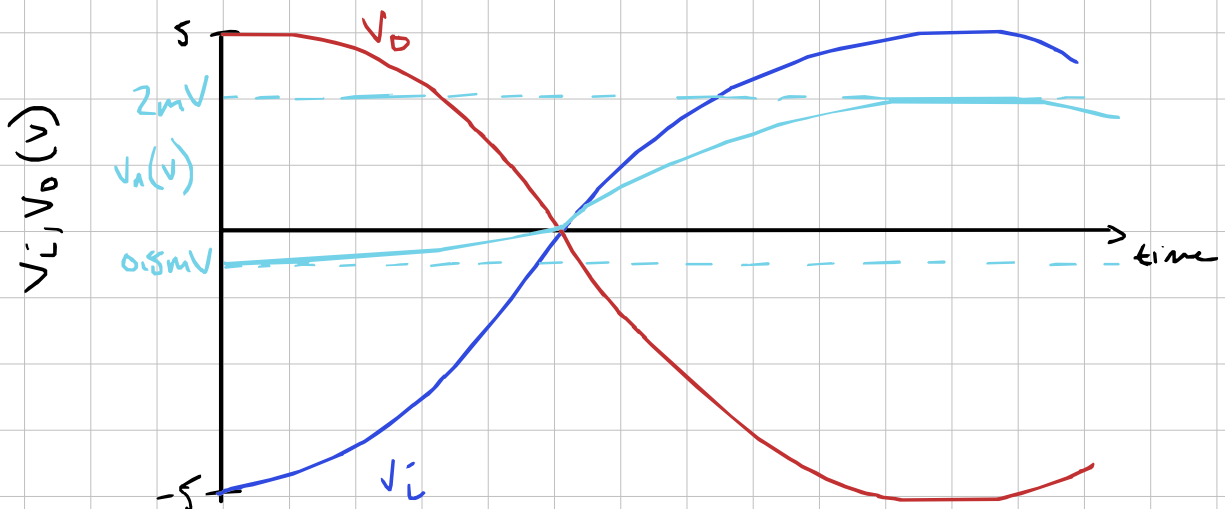
$$\text{peak: } V_i = -5 \text{ V}, V_o = 5 \text{ V}, V_o = -aV_n \Rightarrow V_n = -0.5 \text{ mV}$$

$$V_i > 0, V_o < 0, a = 2.5 \text{ V/mV} = 2500 \text{ V/V}$$

$$L = -0.5(2500)(-1) = 1250$$

$$A = -0.9992 \quad \text{Effectively } -1 \text{ still!}$$

$$\text{peak: } V_i = 5 \text{ V}, V_o = -5 \text{ V}, V_o = -aV_n \Rightarrow V_n = 2 \text{ mV}$$

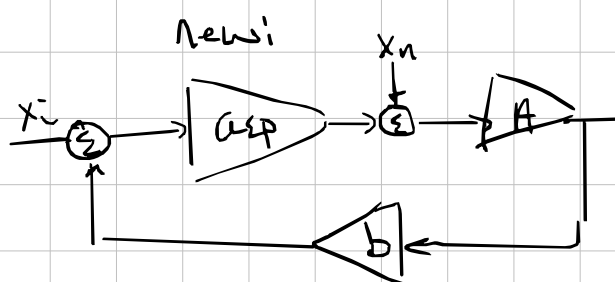
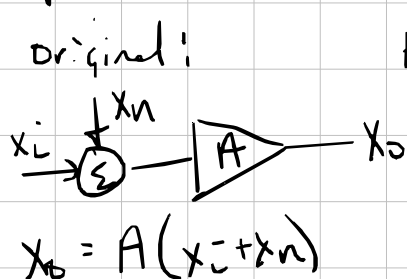


⑦ Book 1.52

This problem can be interpreted in one of two ways:

- ① The pre-amp is added to the power amp with a new feedback added
 - ② The pre-amp is added into the existing feedback loop as part of the power amp
- We will accept both interpretations!

Interpretation ①:



$$x_o = A(x_n + a_{ep}(x_i - bx_o))$$

$$x_o = Aa_{ep}\left(x_i + \frac{x_n}{a_{ep}} - bx_o\right)$$

original: ($x_i=0$)

$$x_o = 2V_{pp} = Ax_n, A=10 \Rightarrow x_n = \frac{2}{10} = 0.2V_{pp}$$

new:

$$x_o = Aa_{ep}(x_i + x_n/a_{ep}) / (1 + Aa_{ep}b)$$

want signal gain to still be 10: ($x_n=0$)

$$\frac{x_o}{x_i} = 10 = a_{ep} A / (1 + Aa_{ep}b), A=10 \quad \textcircled{1}$$

And noise gain to be:

$$x_o = 1mV_{pp}, x_n = 0.2V_{pp} \quad (x_i=0)$$

$$\frac{x_o}{x_n} = 0.005 = A / (1 + Aa_{ep}b) \quad \textcircled{2}$$

putting ② into ①:

$$10 = a_{ep}(0.005) \rightarrow a_{ep} = 2000V/V$$

solving ① with a_{ep} value:

$$10 = 2000 \cdot 10 / (1 + 10(2000)b) \rightarrow b = 0.09995$$

(ideal gain = 10.005)

Interpretation 2

⑦ Book 1.52 Pre-amp Power Amp feedback loop

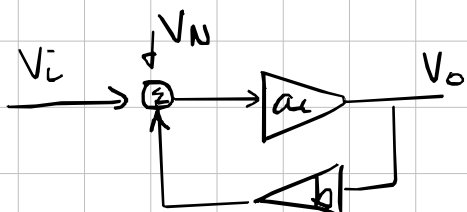
power amp:

$A = 10^4/V$, Output noise: $V_{ON} = 2V_{PP} = 1V_{peak}$

Add pre-amp to reduce hum to $1mV_{PP}$

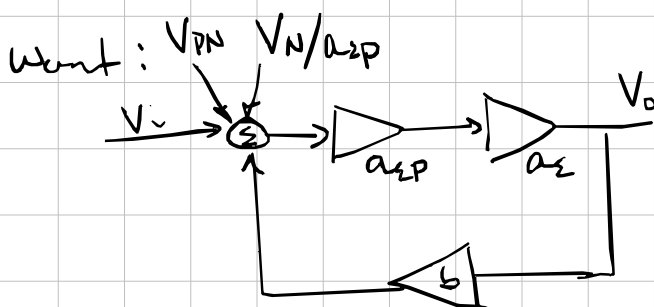
↑ peak-to-peak not clearly stated in problem. Assumed.

Start with:



$V_i \rightarrow$ don't care, assume 0

① $V_{ON} = \frac{a_e}{1 + a_e b} V_N = 2V_{PP} = \frac{1}{b} \frac{1}{1 + 1/L} V_N \quad L = a_e b$



Went: V_{PP} V_N/a_{ep}

$V_i, V_{PP} \rightarrow$ don't care assume 0

② $V_{ON} = \frac{a_{ep} a_e}{1 + a_{ep} a_e b} \left(\frac{V_N}{a_{ep}} \right) = \frac{1}{b} \frac{1}{1 + 1/L} \frac{V_N}{a_{ep}} \quad L = a_{ep} a_e b$

for both cases, $A = 10^4/V$, $\therefore b \approx \frac{1}{10}$

since b is close to 1 $\hat{=}$ a_e , a_{ep} can be assumed to be "large", then:

① $V_{ON} \approx \frac{1}{b} V_N = 2V_{PP} \rightarrow V_N = 0.2V_{PP}$

② $V_{ON} \approx \frac{1}{b} \frac{V_N}{a_{ep}} = 1mV_{PP} \rightarrow a_{ep} = 2000^4/V$

Again, since $a_e \hat{=}$ a_{ep} are both "large" then $L = a_{ep} a_e b$ must be very large, such that $1/(1 + 1/L) \Rightarrow 1$, $\therefore b = 0.1$

(actual gain is ideal)

NOTE: Assumptions on a_e, a_{ep} being large were made. If these values were "small" there would not be enough information to solve this problem. Since it is stated that the original was a "power amplifier" this is a safe assumption.

Of course the results will have slight errors, accordingly.